Sharp Covering-Index Decay for Schatten Classes

Automatic Math Discovery Run `fa_banach_001`

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Status

This packet records a **full result for the Schatten-class covering-index decay problem** left open in Kania–Maślany, *Raja's covering index of L_p spaces*, arXiv:2512.13249. It also records the optimal AUS power exponent of S_p. The result is limited to the classical Schatten classes $S_p(H) = L_p(B(H), \mathrm{Tr})$, not arbitrary non-commutative $L_p(M, \tau)$ spaces.

Source Passage

The source paper notes in Remark 5.3 that for $M = B(H)$ the non-commutative $L_p(M, \tau)$ space is the Schatten class S_p, records the lower bound $\Theta_{S_p}(n) \gtrsim n^{-1/r}$, $r = \min\{p, 2\}$, and then says that upper bounds are not pursued:

> **Proposition 5.2.** *Let (M, τ) be a semifinite von Neumann algebra such that $L_p(M, \tau)$ is infinite-dimensional, let $1 < p < \infty$, and set $r = \min\{p, 2\}$. Then there exists $c_p(M) > 0$ such that*
>
> $$\Theta_{L_p(M,\tau)}(n) \;\geqslant\; c_p(M)\, n^{-1/r} \qquad (n \geqslant 1).$$
>
> Thus the covering index of any non-commutative L_p-space decays at least like $n^{-1/r}$ with $r = \min\{p, 2\}$.
>
> *Remark* 5.3. If $M = L_\infty(\Omega, \mu)$ and $\tau(f) = \int_\Omega f\, d\mu$, then $L_p(M, \tau)$ is just $L_p(\mu)$ and Proposition 5.2 yields a lower bound of order $n^{-1/r}$ with $r = \min\{p, 2\}$. Our explicit covering in the commutative case (Proposition 3.1) shows that the optimal exponent is at most $1/p$; when $L_p(\mu)$ moreover admits an equivalent p-AUS norm, Corollary 3.3 gives $\Theta_{L_p(\mu)}(n) \asymp n^{-1/p}$. Thus the abstract non-commutative argument is not sharp even in the commutative setting when $p > 2$.
>
> If $M = B(H)$ with the canonical trace, then $L_p(M, \tau)$ is the Schatten class S_p on H. Proposition 5.2 then shows that
>
> $$\Theta_{S_p}(n) \;\gtrsim\; n^{-1/r}, \qquad r = \min\{p, 2\}.$$
>
> We do not pursue corresponding upper bounds in the non-commutative setting here.

The appendix then states the remaining sharp-decay problem explicitly:

and hence $L_p(M,\tau)$ is asymptotically uniformly smooth of power type $r>1$. In particular, the natural norm on $L_p(M,\tau)$ is r-AUS with $r=\min\{p,2\}$. This is the input used in the proof of Proposition 5.2: once $L_p(M,\tau)$ is r-AUS, Raja's general theorem (Theorem 1.6) yields a power-type lower bound

$$\Theta_{L_p(M,\tau)}(n) \;\gtrsim\; n^{-1/r}.$$

Determining the *precise* optimal AUS exponent and the corresponding sharp covering-index decay rate for specific non-commutative L_p-spaces remains an interesting open problem, even for classical examples such as the Schatten classes $S_p = L_p(B(H),\mathrm{Tr})$.

Definitions

For a Banach space X, the essential inradius of a set $A \subseteq X$ is

$$\varrho(A) = \sup\{r>0 : \exists x \in A,\ \exists Y \leq X \text{ finite codimensional},\ x + r(B_X \cap Y) \subseteq A\}.$$

Raja's covering index is

$$\Theta_X(n) = \inf\left\{ \max_{1\le k\le n} \varrho(A_k) : B_X = \bigcup_{k=1}^{n} A_k,\ A_k \text{ closed, convex, bounded} \right\}.$$

Let H be an infinite-dimensional Hilbert space. The Schatten class $S_p(H)$, $1 \le p < \infty$, is the space of compact operators T on H whose singular-value sequence belongs to ℓ_p, with norm $\|T\|_p = (\mathrm{Tr}\,|T|^p)^{1/p}$.

Proof Intuition

There are two complementary mechanisms. The upper bound is a matrix version of the scalar L_p block cover from the source paper. Split

$$H = H_1 \oplus \cdots \oplus H_n$$

and cover the unit ball according to which diagonal corner $P_k T P_k$ has small S_p-norm. Pinching onto diagonal corners is contractive, so one corner must have norm at most $n^{-1/p}$. The finite-codimensional inradius test is defeated by placing a rank-one operator far out in that corner, chosen to annihilate the finitely many functionals defining the subspace.

The lower bound is shorter. The diagonal operators form a 1-complemented copy of ℓ_p inside $S_p(H)$. The source paper records that covering index is monotone for contractively complemented subspaces, and it also gives the sharp scalar estimate $\Theta_{\ell_p}(n) \asymp n^{-1/p}$ for $1<p<\infty$. Hence $S_p(H)$ cannot have faster covering-index decay than the diagonal ℓ_p part.

Sharp Covering-Index Decay

Theorem 1. *Let H be an infinite-dimensional Hilbert space and let $1 \le p < \infty$. Then for every $n \in \mathbb{N}$,*

$$\Theta_{S_p(H)}(n) \le n^{-1/p}.$$

Indeed, $B_{S_p(H)}$ has a cover by n closed convex sets, each of essential inradius exactly $n^{-1/p}$.

Proof. Fix $n \in \mathbb{N}$, and write

$$H = H_1 \oplus \cdots \oplus H_n$$

as an orthogonal direct sum of infinite-dimensional closed subspaces. Let P_k be the orthogonal projection onto H_k, and define

$$C_k(T) = P_k T P_k, \qquad T \in S_p(H).$$

Put $a = n^{-1/p}$, and define

$$A_k = \{T \in B_{S_p(H)} : \|C_k(T)\|_p \le a\} \qquad (1 \le k \le n).$$

Each A_k is closed, convex, and bounded.

Let $E(T) = \sum_{k=1}^n P_k T P_k$ be the finite pinching. If $U_\varepsilon = \sum_{k=1}^n \varepsilon_k P_k$ for $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{-1,1\}^n$, then

$$E(T) = 2^{-n} \sum_{\varepsilon \in \{-1,1\}^n} U_\varepsilon T U_\varepsilon.$$

Since the Schatten norm is unitarily invariant and convex, $\|E(T)\|_p \le \|T\|_p$. Since $E(T)$ is block diagonal,

$$\|E(T)\|_p^p = \sum_{k=1}^n \|C_k(T)\|_p^p.$$

Thus if $\|T\|_p \le 1$, at least one $\|C_k(T)\|_p$ is at most $n^{-1/p}$, and $B_{S_p(H)} = \bigcup_k A_k$.

Also $aB_{S_p(H)} \subseteq A_k$, so $\varrho(A_k) \ge a$.

We prove the reverse inequality. Fix k. Suppose that $T \in A_k$, $Y \le S_p(H)$ is finite codimensional, and $r > a$. We shall find $W \in B_{S_p(H)} \cap Y$ such that $T + rW \notin A_k$. Set $V = C_k(T)$, so $\|V\|_p \le a$.

Choose functionals $\Phi_1, \dots, \Phi_m \in S_p(H)^*$ with

$$Y = \bigcap_{j=1}^m \ker \Phi_j.$$

By the standard trace-duality representation of Schatten functionals, and using $S_1(H)^* = B(H)$ when $p = 1$, there are operators $B_j \in B(H)$ such that for rank-one operators $\theta_{u,v}(\xi) = \langle \xi, v\rangle u$ one has

$$\Phi_j(\theta_{u,v}) = \langle B_j u, v\rangle.$$

Let $R \le P_k$ be a finite-rank orthogonal projection on H_k. Since $V \in S_p(H_k)$, finite-rank compressions approximate V in S_p-norm; hence R may be chosen so that

$$\big(\|RVR\|_p^p + r^p\big)^{1/p} - \|V - RVR\|_p > a. \tag{1}$$

Indeed, the left side tends to $(\|V\|_p^p + r^p)^{1/p} > r > a$.

The subspace $K = H_k \ominus R(H_k)$ is infinite-dimensional. Pick a unit vector $u \in K$, then pick a unit vector

$$v \in K \cap \operatorname{span}\{B_1 u, \dots, B_m u\}^\perp.$$

This is possible because only finitely many vectors are excluded. Put $W = \theta_{u,v}$. Then W is supported in the k-th corner, $\|W\|_p = 1$, and $\Phi_j(W) = 0$ for every j, so $W \in Y$.

Since RVR acts on $R(H_k)$ and W acts from K to K, the operator $RVR + rW$ is block diagonal. Hence

$$\|RVR + rW\|_p^p = \|RVR\|_p^p + r^p.$$

Using (1),

$$\|C_k(T + rW)\|_p = \|V + rW\|_p \geq \|RVR + rW\|_p - \|V - RVR\|_p > a.$$

Thus $T + rW \notin A_k$. Therefore no finite-codimensional ball of radius larger than a is contained in A_k, and $\varrho(A_k) \leq a$. Since $\varrho(A_k) \geq a$, the covering has pieces of essential inradius exactly a, proving the theorem. $\square$

Theorem 2. *Let H be an infinite-dimensional Hilbert space and let $1 < p < \infty$. Then*

$$\Theta_{S_p(H)}(n) \asymp n^{-1/p}.$$

Proof. The upper bound is Theorem 1. For the lower bound, choose an orthonormal sequence (e_j) in H, and let $D \subset S_p(H)$ be the diagonal subspace

$$D = \left\{ \sum_j a_j \, e_j \otimes e_j : (a_j) \in \ell_p \right\}.$$

Then D is isometric to ℓ_p. The diagonal conditional expectation

$$\mathcal{E}(T) = \sum_j \langle Te_j, e_j \rangle \, e_j \otimes e_j$$

is contractive on $S_p(H)$; equivalently, it is the norm-one conditional expectation onto the diagonal masa, and it follows by finite pinching and approximation. Thus D is a contractively complemented subspace of $S_p(H)$.

Kania–Maślany's monotonicity lemma for contractively complemented subspaces [1, Lemma 3.3] gives

$$\Theta_{\ell_p}(n) = \Theta_D(n) \leq \Theta_{S_p(H)}(n).$$

Their scalar L_p estimate, applied to the counting measure, gives $\Theta_{\ell_p}(n) \asymp n^{-1/p}$ for $1 < p < \infty$ [1, Corollary 3.4 and the preceding remark]. Therefore

$$\Theta_{S_p(H)}(n) \gtrsim n^{-1/p}.$$

Together with the upper bound, this proves the claimed sharp decay. $\square$

Remark 1. *The upper-bound theorem also holds for $p = 1$. The lower-bound argument reduces the $p = 1$ endpoint to the known scalar value of Θ_{ℓ_1}; the source paper's non-commutative lower-bound discussion, however, is stated for $1 < p < \infty$, so the main sharp-decay statement above is formulated in that same range.*

Optimal AUS Exponent for S_p

Proposition 1. *Let $1 < p < \infty$. The largest AUS power exponent obtainable by an equivalent norm on $S_p(H)$ is*

$$\min\{p, 2\}.$$

Proof. The source paper recalls from the non-commutative Clarkson inequalities that the usual Schatten norm is r-AUS for $r = \min\{p, 2\}$ [1, Appendix]. Hence the optimal exponent is at least r.

It remains to exclude any $q > r$. AUSability passes to subspaces. If $1 < p \leq 2$ and $S_p(H)$ admitted an equivalent q-AUS norm for some $q > p$, then the diagonal subspace $D \cong \ell_p$ would be q-AUSable. Raja's lower bound, quoted as Theorem 1.6 in the source paper, would imply

$$\Theta_{\ell_p}(n) \gtrsim n^{-1/q}.$$

This contradicts the scalar estimate $\Theta_{\ell_p}(n) \asymp n^{-1/p}$, because $q > p$.

If $p \geq 2$, fix a unit vector $e_0 \in H$. The row space

$$R = \{\theta_{e_0,u} : u \in H\} \subset S_p(H)$$

is isometric to the Hilbert space H, since each $\theta_{e_0,u}$ has the single singular value $\|u\|$. If $S_p(H)$ were q-AUSable for some $q > 2$, then H would be q-AUSable. Raja's lower bound would give $\Theta_H(n) \gtrsim n^{-1/q}$, contradicting Kania–Maślany's exact Hilbert computation $\Theta_H(n) = n^{-1/2}$ [1, Theorem 2.2].

Thus no exponent larger than $\min\{p, 2\}$ is possible. ☐

What This Solves and What It Does Not

For classical Schatten classes, the source paper's open Schatten covering decay question is resolved: the sharp power is $1/p$, i.e.

$$\Theta_{S_p(H)}(n) \asymp n^{-1/p} \qquad (1 < p < \infty).$$

The packet also gives the precise optimal AUS power exponent, $\min\{p, 2\}$. In particular, for $p > 2$, the sharp covering-index decay is governed by the diagonal ℓ_p subspace rather than by the optimal AUS exponent 2.

The packet does not settle arbitrary non-commutative $L_p(M, \tau)$ spaces. The upper-bound proof uses the concrete decomposition of H into matrix corners, and the lower-bound proof uses the diagonal ℓ_p masa in $B(H)$.

Verification and Novelty Check

The proof is abstract and uses no numerical computation. The code `code/make_open_problem_crops.py` renders the two source-passage crops from `source_paper.pdf`.

Local registry, solution, attempt, and proof-gap indexes had only the earlier partial packet from this run for arXiv:2512.13249, which has now been promoted and rewritten here. A bounded web/arXiv search on June 16, 2026 for "Raja's covering index Schatten $n^{-1/p}$", "Theta S_p covering index", "Schatten Raja covering index ell_p", and "essential inradius S_p Theta" found Raja's original paper arXiv:2503.03721 and the source paper arXiv:2512.13249, but no later paper giving the diagonal lower-bound upgrade or the resulting sharp Schatten decay. The source paper itself says that non-commutative upper bounds are not pursued and that the sharp Schatten-class decay remains open.

Revision History

The first pass on this target produced only the upper bound $\Theta_{S_p(H)}(n) \leq n^{-1/p}$. That superseded partial packet is kept inside this packet at `history/partial_upper_bound/`. The present version adds the diagonal ℓ_p lower bound and the optimal AUS exponent argument, so the indexed durable result is this full packet.

Human Review Recommendation

Review as a likely valid full solution of the Schatten covering-index decay subproblem. The main checks are the finite-codimensional rank-one tail argument in Theorem 1, the use of the diagonal conditional expectation and complemented-subspace monotonicity in Theorem 2, and the subspace obstruction in Proposition 1.

References

[1] T. Kania and N. Maślany, *Raja's covering index of L_p spaces*, arXiv:2512.13249.

[2] M. Raja, *A covering index for Banach spaces*, arXiv:2503.03721.